Chapter 1

Generative AI as a Facilitator of Deliberate Practice in Translator Training

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Advancement along an expertise trajectory, whether in training or professional contexts, stems from translators having ample opportunity to engage in deliberate practice, defined in the expertise studies literature as strategically designed activities to improve performance (Ericsson, **KrampeTesch-Römer1993**). In order for practice to be deliberate, a number of core conditions need to be met, including, among others, intrinsic learner motivation, focused and self-directed performance monitoring, informative (and relatively immediate) feedback, tasks being structured and undertaken at an appropriate difficulty level, and learners having opportunities to correct errors (**Shreve2006**). In translator training contexts, complex, heterogeneous learner profiles, particularly when it comes to knowledge, skills, and competences (**EMT2022**), pose inherent challenges in attempts to make sure these conditions are adequately met. In addressing these challenges, and in optimizing opportunities for deliberate practice in general, Generative AI offers pedagogical value as a vehicle to guide self-directed learning. This chapter will discuss how generative AI can be used to facilitate deliberate practice in translator training, where students engage in self-directed metacognitive activity to reflect on and assess facets of their own performance, and, in doing so, acquire and advance their expertise.

1 Introduction

Technological advancement, most recently in the realm of generative AI, is rapidly changing the roles and responsibilities of professional translators. It is

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also re-shaping the competences they need to possess to find success in the language industry, as reflected in the recently updated EMT Translation Competence Framework (**EMT2022**). In times of seemingly perpetual change and a corresponding need to adapt (**Angelone2022**), one aspect regarding competence has remained fairly constant, namely the need for translators to critically reflect on, assess, and optimize their own performance. This capacity is articulated in several of the sub-competences pertaining to overarching 'translation competence' in the EMT framework, such as translators being able to "analyse a source document, identify potential textual and cognitive difficulties and assess the strategies and resources needed to reformulate it", and "analyse and justify their translation solutions and choices" (**EMT2022**). It is also manifest in the domain of personal and interpersonal competence as the ability to "manage workload, cognitive load, stress and critical professional situations", and "continuously self-evaluate, update and develop competences and skills through personal strategies and collaborative learning" (**EMT2022**).

Competence, which can broadly be defined as the set of knowledge and skills needed to successfully translate, is closely tied to expertise, which, depending on research paradigm and theoretical foundation, can be described as 'consistently superior performance' in a given task domain (**EricssonCharness1994**), or 'optimal performance' across varied, but interrelated task domains (**HatanoInagaki1986**). Performance, when it comes to successful translation, involves not only quality (such as identifying and mitigating errors or successfully adhering to established project specifications), but also productivity (such as leveraging assistive technologies to expedite output or utilizing ergonomically sound approaches when facing time constraints). Expertise should be regarded as something developmental and incremental (**Shreve2018**) rather than a desirable 'end state' in the sense of an 'expert' translator. With this in mind, training for translation expertise acquisition and advancement has an important place from the outset.

In order to advance along an expertise trajectory, for example from 'novice', to 'advanced beginner', to 'competent', to 'proficient', to 'expert' (**DreyfusDreyfus1986**), experience alone will likely not suffice. Instead, translators should intentionally seek opportunities to engage in 'deliberate practice', which can be defined as "individualized training activities especially designed by a coach or teacher to improve specific aspects of an individual's performance through repetition and successive refinement" (**EricssonLehmann1996**:278-279). Deliberate practice necessitates moving beyond "plateaus where one is comfortable and confident" (**HornMasunaga2006**:601). Unlike the somewhat predictable activity and approaches that translators might be inclined to encounter in their daily work, deliberate practice involves "exploring alternative methods with unknown reliability" (Ericsson, **KrampeTesch-Römer1993**:368). Indeed, common industry constraints on time, budget, infrastructure, and risk-taking in general make deliberate practice something distinct from 'work'. It is more of an external, upfront investment that can ultimately optimize work performance.

2 2 Core conditions of deliberate practice

In order for practice to be deliberate, a number of important core conditions need to be met. Translation Studies scholars have outlined these as they pertain to the task of translation (**Shreve2006**). Perhaps most important is the condition of translators receiving immediate, informative feedback on their performance. "In the absence of adequate feedback, efficient learning is impossible and improvement minimal" (Ericsson, **KrampeTesch-Römer1993**:367). Informative feedback ideally encompasses aspects of translation that are transferable and applicable across tasks. For example, feedback on the efficacy or inefficacy of a given external resource or assistive technology is likely more informative than feedback on a misspelled word that the translator might encounter in the context of one given translation but never again.

As a second condition, translators need to have opportunities to correct errors. In pedagogical contexts, this condition is often met through workshopping, where students submit and receive feedback on drafts, followed by the opportunity to submit revisions in which errors brought to their attention can be corrected. Learning is particularly enhanced when error correction stems from discovery-based learning, where errors are annotated, but deliberately not fully spelled out by trainers. The very presence of a trainer is yet another core condition of deliberate practice. The trainer, or 'coach', is responsible not only for providing informative feedback, but also for establishing individualized learning objectives (**MillerEtAl2020**) and designing tasks based on firm understanding of the learner's pre-existing knowledge. Some proponents of deliberate practice-oriented training suggest that individualized supervision is ultimately preferred over group-based instruction (Ericsson, **KrampeTesch-Römer1993**:367).

The trainer also plays a pivotal role in fulfilling another core dimension of deliberate practice, namely making sure the translator is engaging in tasks of an appropriate difficulty level. Stagnation, due to strict adherence to the tried and true, can result in a plateau effect that stands in the way of expertise advancement. Translators may encounter this when working on tasks of a uniform difficulty level (same source text length, same readability level, same set of external resources, same project specifications, same assessment parameters, etc.).

Trainers can push trainees to work outside of their comfort zones and help determine "when transitions to more complex and challenging tasks are appropriate" (Ericsson, **KrampeTesch-Römer1993**:367).

Deliberate practice hinges on the learner's intrinsic motivation, another core condition. Translators looking to advance along an expertise trajectory need to embrace working outside of their comfort zones and have a firm belief that deliberate practice will ultimately improve their performance. Deliberate practice takes immense time and effort. The oft-debated 10,000-hour 'rule' to becoming an 'expert' (**Gladwell2011**) is often cited in attempts to quantify just how much time and effort are required. This would amount to "two and a half years of sustained effortful practice every day for five hours" (**Shreve2019**), raising questions regarding feasibility of this 'rule' and its place in the deliberate practice model.

Deliberate practice calls for the learner's commitment to "conscious performance monitoring" (**HornMasunaga2006**:601) and engaging in full concentration, as opposed to "mindless, routine performance" (**Ericsson2006**). This takes self-discipline and learner dedication to honing metacognitive capacities. Strategic scaffolding by the trainer, along with trainer/trainee dedication to the central ideas of cognitive constructivism (**Piaget1952**), are instrumental in bolstering learner metacognition and performance monitoring processes.

Figure **??** provides an overview of the core conditions of deliberate practice outlined in this chapter. It does not represent an exhaustive list of all conditions mentioned in the Expertise Studies literature, but rather focuses on those conditions taken up in Translation Studies to date.

3 Challenges to the acquisition of deliberate practice

It is worth noting here that empirical research on the benefits of deliberate practice on translation performance is still quite scant. This dearth can at least be partly explained by a series of inherent challenges associated with its implementation in formal training contexts. As mentioned in the previous section, several proponents of deliberate practice suggest that its benefits are best realized in contexts involving learners working one on one with individual trainers and in the absence of a pre-set curriculum (Ericsson, **KrampeTesch-Römer1993**:367). This type of design does not readily align with the fashion in which translators are usually trained for a number of different reasons, starting with the financial constraint of needing to hire a personal trainer.

Another constraint potentially standing in the way of meeting several of the core conditions of deliberate practice is the degree of learner heterogeneity commonly found in classroom-based training contexts. Students of translation often

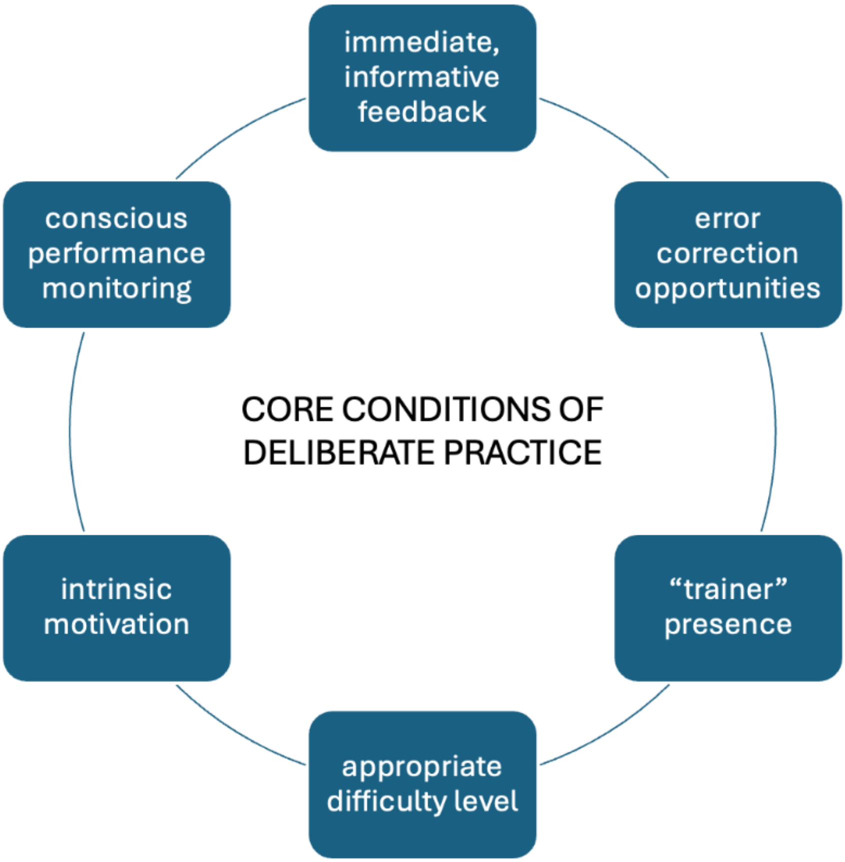

Figure 1: Core conditions of deliberate practice

have widely varying levels of competence and experience, not to mention diverse learning needs and interests. This can make it difficult for trainers to establish truly individualized learning objectives and corresponding tasks to meet them. The problem is exacerbated when student enrollments are high. Feedback, arguably the most important dimension of deliberate practice, often becomes less detailed and, out of necessity, much less immediate. Peer feedback and self-feedback activities can help address this gap, but immediacy, as a criterion for practice to be deliberate, often remains very difficult to obtain.

Beyond time constraints, the deliberate practice condition of informative, immediate feedback might not be met if the kind of feedback being provided is not transferable in the sense of applying to future translation tasks. Oftentimes, assessment rubrics are used to mark up errors in accordance with various textual levels, such as grammar, word choice, and syntax. If the feedback given pertains to patterns along these lines that are applicable across tasks, it could be regarded as truly informative. If, on the other hand, feedback simply consists of marking up one-off errors pertaining to items that the translator may never encounter again in future translations, such as an isolated collocation error, it is relatively shallow and not particularly informative.

Learner heterogeneity in large cohort settings also presents challenges when it comes to meeting the deliberate practice condition of making sure tasks are being undertaken at an appropriate level of difficulty. Translation practice courses are often informed by trainer intuition and prediction of appropriate difficulty level, in turn based on such facets as level of study, in-take examination results, various cognitive process metrics (**SunShreve2014**), or previous student performance and corresponding 'rich points' (PACTE **Group2011**). However, studies have shown a danger for misalignment between perceived or predicted problems and what actually proves to be problematic when translating (**Angelone2018**). What is assumed to be difficult actually is not necessarily, making attempts at predicting and setting an appropriate difficulty level challenging, particularly across a wide range of students with diverging needs. Furthermore, the idea of having students work outside of their comfort zone, or at the periphery of what they can realistically accomplish, for purposes of advancing expertise may contradict approaches to translator training that embrace the predictability of staying within the learner's comfort zone.

According to deliberate practice guidelines, the learner's intrinsic motivation needs to be constant. Having translation students work outside of their comfort zone in pedagogical contexts, particularly when grades are involved, runs the risk of hampering such motivation. The outcomes of deliberate practice would need to be gauged using metrics beyond formal grades, with a focus on helping learners become more self-reflective translators. Intrinsic motivation would come not so much from getting good grades or doing well in a given course, but rather from seeing the benefits of putting in hard work and embracing difficulty for purposes of becoming better translators.

4 4 Generative AI as a facilitator of deliberate practice

Against the backdrop of the conditions of deliberate practice presented in Section **??** and the constraints potentially standing in its way, as outlined in Section **??**, the question remains: how can we best go about facilitating deliberate practice in translator training for purposes of expertise acquisition and advancement? In particular, how can we establish the requisite highly individualized, 'coach'-oriented approach at the heart of deliberate practice? The pedagogical features of generative AI, in providing real-time feedback and prompt-driven interaction in a user-centered fashion, would seem to hold potential in this regard, as will be illustrated through a series of concrete scenarios in Section **??**.

At the time of writing, we are still witnessing the dawn of generative AI as a vehicle for optimizing translation, both in professional and pedagogical contexts. Its increasingly ubiquitous integration by LSP companies in project workflows has drawn attention to a need for artificial intelligence (AI) literacy (**Krüger2023**), alongside MT literacy and data literacy in a broad sense. AI literacy can be defined as "a set of competencies that enables individuals to critically evaluate AI technologies, communicate and collaborate effectively with AI, and use AI as a tool online, at home, and in the workplace" (**LongMagerko2020**:2). In conjunction with discussions of translator performance, much of the discourse on AI intelligence to date has focused on how generative AI can be leveraged to facilitate human-in-the-loop translation, with an emphasis on the translation product. Outside of several of the chapters in this volume, relatively little discussion has been dedicated to the potential benefits of generative AI as a conversational agent, focusing less on generating translated content, and more so on enabling translators to reflect on their performance and engage in translation tasks to facilitate deliberate practice.

Extending on sociocultural learning theories (**Vygotsky1986**), some have come to regard a generative AI tool like ChatGPT as a 'more knowledgeable other' (MKO), in essence taking on the role of a personalized trainer that can "lead the learner from the zone of current development to the zone of proximal development - the space where one cannot quite master a content/task of their own, but they can with the help of an expert" (**Stojanov2023**). As an MKO ChatGPT can address the aforementioned conditions of deliberate practice, particularly the presence of a one-on-one personalized trainer. Through real-time responses to learner prompts, ChatGPT ensures the immediacy of feedback that is so difficult to obtain in a larger enrollment classroom-based translator training environment. The learner's intrinsic motivation is likely to be heightened when training is personalized and self-driven, rooted in immediate, informative feedback, and interactive, driven by their own prompts in relation to aspects of their own performance.

5 5 Generative AI as a scaffold for self-directed learning

The utilization of generative AI for purposes of deliberate practice calls for the learner to partake in self-directed learning (SDL), where "individuals take the initiative, with or without the help of others, in diagnosing their learning needs, setting learning goals, identifying resources, choosing appropriate learning strategies, and evaluating their learning outcomes" (**Knowles1975**). In this case, generative AI helps scaffold learning in line with learner prompts. Translation trainers well-versed in the conditions of deliberate practice can provide learners with valuable information on the nature of prompts they should enter. However, self-directed learners need to be "able, ready, and willing to prepare, execute and complete learning independently" (**JossbergerEtAl2010**:419). The need for learner independence does not make the trainer superfluous, but rather shifts the focus of assignments undertaken and how they are assessed.

Models of self-directed learning bear very close resemblance to the deliberate practice model. One such model that is widely cited in the literature consists of three closely interrelated dimensions: 1) self-management, 2) self-monitoring, and 3) motivation (**Garrison1997**). Self-management involves the learner establishing concrete learning goals and managing learning resources to achieve these goals. In other words, they take control, deciding on the tasks in which they will engage. From a deliberate practice perspective, through strategic prompts, the translator can leverage generative AI to annotate errors in their translations. Generated annotations could then serve as a framework for the translator to self-discover the nature of the errors. Generative AI could then be prompted to and provide similar translation tasks, with the goal of engaging the learner in deliberate practice centered around a certain error pattern (such as avoiding false cognates, erroneous literal translation, or problematic translationese at a syntactic level). Beyond error detection and mitigation, the translator can also use generative AI prompting to self-manage the difficulty level of the tasks they are undertaking. Section **??** provides more concrete scenarios and descriptions along these lines.

Self-monitoring, the second component of Garrison's model, pertains to the learner's metacognitive processes. As an important dimension of self-directed learning, self-monitoring "requires learners to take responsibility to construct meanings" (**Garrison1997**). Through interactive feedback, generative AI can shed valuable light to help learners identify salient features of the translation task on which to focus their attention. For example, translators can enter a prompt asking generative AI to annotate source content that could be anticipated or predicted to present challenges in translation. Over the past decade, screen recording has found a place in process-oriented translator training for purposes of fostering self-monitoring and to enhance learner metacognition based on documentation of translation behaviors suggesting problems, including pausing, information retrieval, and revision (**Angelone2019**). At present, generative AI tools like ChatGPT do not offer functionality where translators can upload screen recordings of their work for purposes of receiving analytic feedback at a granular level. Given recent advancements in this technology however, such as APIs that can provide automated video summarization, it is quite likely that utilization of generative AI tools for such purposes is not too far away.

The third component of Garrison's self-directed learning model, motivation, directly aligns with motivation as a core condition of deliberate practice. Whereas deliberate practice regards motivation as the learner embracing challenge through an inherent desire to become better, Garrison draws attention to the importance of motivation for purposes of staying on task. The conversational interface of generative AI tools such as ChatGPT requires active participation on behalf of the learner, and, thereby, a heightened need to stay on task. However, without the physical presence of actual trainers or peers, staying on task, and motivation in general, is not a given. ChatGPT is not inclined to openly praise the performance of the learner to pique interest and motivation. Indeed, at present, the ChatGPT interface itself is quite basic, lacking any structural or discourse elements that might lend themselves well to inherently facilitating learner motivation. It will be interesting to see if this changes over time, perhaps in line with empirical user experience studies.

6 6 Application scenarios

This chapter will now provide a series of concrete scenarios to illustrate how generative AI can be used to facilitate each of the core dimensions of deliberate practice put forward in Section ?? and to help translators advance along an expertise trajectory. The examples will be based on interaction with ChatGPT based on GPT-4o,[1] given the relatively ubiquity and popularity of this particular generative AI tool at the time of writing. This focus on ChatGPT for illustrative purposes does not discredit the growing range of other generative AI tools available for use in a similar fashion. The translation scenario being used for purposes of contextualization is the German-English translation of web content from a German private health insurance company,[2] translated for an international, English-speaking audience for informative purposes. The English translation was generated using DeepL.[3]

Figure ?? provides side-by-side alignment of the source and target content.

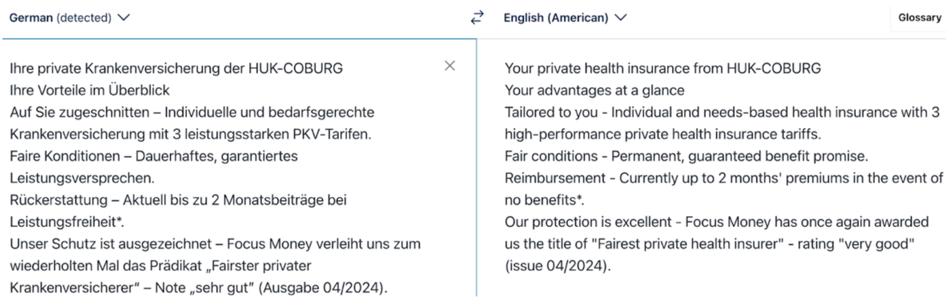

Figure 2: German-English translation of private health insurance content

As a point of departure, and to underscore the importance of strategic prompting when it comes to utilizing Gen AI for facilitating deliberate practice, Table 1 depicts the relatively shallow output proposed by ChatGPT based on the following generic prompt:

"Set up translation training tasks with the goal of facilitating deliberate practice".

Table 2 depicts corresponding training tasks proposed by ChatGPT in parallel to the content in Table 1.

While tapping into the core conditions of deliberate practice put forward in the literature, this generic framework proposed by ChatGPT seems to focus first and foremost on group-based (rather than truly personalized) training and also seems to make the assumption that learners are starting 'from scratch' (moving

[1] https://openai.com/index/gpt-4/

[2] https://www.huk.de/gesundheit-vorsorge-vermoegen/krankenversicherung/krankenvollversicherung.html#h

[3] https://www.deepl.com/en/translator

Table 1: : Chat GPT output for setting up translation tasks to facilitate deliberate practice

Setting up translation training tasks for deliberate practice involves focusing on specific skills, receiving immediate feedback, and gradually increasing difficulty.

1. Identify key skills and goals

Skills: Vocabulary, grammar, idiomatic expressions, cultural context, speed, accuracy, style, and tone.

Goals: Improve comprehension, enhance translation quality, increase speed without losing accuracy, and develop a nuanced understanding of both source and target languages.

2. Feedback and reflection

Immediate Feedback: Provide prompt feedback on translations, highlighting errors and explaining corrections.

Self-Assessment: Encourage learners to reflect on their work and identify areas for improvement.

3. Deliberate practice cycle

Plan: Identify specific skills to focus on in each session.

Practice: Perform targeted translation tasks.

Feedback: Receive and review feedback from instructors or peers.

Reflect: Reflect on the feedback and identify areas for improvement.

Repeat: Repeat the cycle with new tasks, gradually increasing difficulty.

Table 2: : ChatGPT training task recommendations

Week 1: Basic sentences

Translate 10 simple sentences from the source language to the target language, with a focus on vocabulary and basic grammar.

Week 2: Themed paragraphs

Translate a paragraph related to a specific theme (e.g., technology), with a focus on thematic vocabulary and coherence.

Week 3: Idiomatic expressions

Translate sentences rich in idiomatic expressions, with a focus on cultural nuances and idiomatic accuracy.

Week 4: Time translations

Translate a 200-word article within 20 minutes, with a focus on speed and maintaining accuracy under time constraints.

Week 5: Reverse translation

Perform a back translation exercise, with a focus on understanding equivalence and identifying subtle changes in meaning.

from basic vocabulary to field-specific terminology, to cultural and idiomatic expressions, for example). Given the inherent learner heterogeneity described in Section 3, many translators would not directly benefit from such a lockstep, linear approach. As far as difficulty is concerned, it seems as if there is less focus on working at an appropriate difficulty level per se and more of an emphasis on incorporating a variety of potential sources of difficulty in a more fragmented sense (thematic vocabulary, idiomatic accuracy, time pressure, etc.). This is not to suggest that ChatGPT cannot personalize deliberate practice and cater to the needs of the individual user, but rather highlights the important, active role that the individual learner needs to play in order to benefit from generative AI. This starts with utilizing more granular prompts in accordance with each specific condition of deliberate practice rather than taking a broad-brush approach such as that associated with the generic prompt that generated the output here.

6.1 Immediate, informative feedback

At a very basic level, translators can utilize ChatGPT to get immediate, informative feedback on their performance by entering source content and their corresponding translations, along with correlating project specifications for the task (such as audience and purpose), in conjunction with the following prompt:

"Provide me with feedback on the following translation, German to English, of content from a private health insurance company in Germany for an international English-speaking audience, with the purpose of providing information".

Table 3 depicts the feedback obtained from ChatGPT in relation to this prompt.

Table 4 provides an overview of the response provided by ChatGPT, illustrating how feedback can become truly informative and transferable, as opposed to isolated to the one translation task at hand.

Interestingly, ChatGPT seems to attribute transferability to each piece of general feedback provided in response to the initial prompt, as depicted in Table 3. This suggests a potential tendency for ChatGPT to conflate task-specific and transferable feedback. However, this did not hold true in the context of a different simulated translation task, where some of the task-specific feedback was deliberately filtered out and not inherently regarded by ChatGPT as also being transferable. Nevertheless, from a prompting standpoint, for purposes of obtaining truly informative feedback, the translator may want to ask: "Which of this feedback is likely specific only to this one translation task and which of the feedback is transferable across translations in the same domain?", with the latter being more beneficial from a deliberate practice perspective.

Table 3: : ChatGPT feedback on a translation

Your translation is clear and accurate, but I have a few suggestions to make it sound more natural for an international English-speaking audience. Here are the changes made:
1.Changed ”advantages” to ”benefits” for a more common term in health insurance.
2. Changed ”tariffs” to ”plans” to better fit the context.
3. Added a colon after ”Your benefits at a glance” for better readability.
4. Capitalized ”Private Health Insurer” for consistency with the title awarded.

This feedback was generated in a matter of a few seconds, overcoming the time constraint often standing in the way of feedback being immediate in a typical translator training environment. As mentioned in Section **??**, feedback becomes truly informative when it is applicable across translation tasks under similar conditions. With this in mind, the translator could enter the following prompt in response to the obtained feedback:
“Which of this feedback would be applicable when translating other private health insurance content from German into English in the future”?

Table 4: : Informative, transferable feedback

When translating content for other private health insurance companies in the future, you can apply the following feedback to ensure clarity, accuracy, and appropriateness for an international English-speaking audience:
1. Use Clear and Common Terms:
Replace terms like ”advantages” with ”benefits” for more common understanding.
2. Ensure Readability and Natural Flow:
Add colons or other punctuation to improve readability (e.g., ”Your benefits at a glance:”).
3. Maintain Consistent Formatting:
Use bullet points to list benefits for easy reading and better presentation.

6.1 6.2 Error correction opportunities

Immediate, informative feedback, as a core condition of deliberate practice, provides the translator with granular insight into the nature of errors. In hopes of advancing along an expertise trajectory, it is just as important for the translator to have opportunities to correct errors they have made by virtue of having but not spell out, and to translate additional source content in which they would be prone to making similar errors. ChatGPT can be prompted to annotate translation errors without correcting or describing them, using the same approach outlined in the preceding subsection. For this purpose, the following prompt could be used:

"Annotate errors in the following English translation of German source content for a private health insurance company using a bold font. The translation has an informative function and targets an international audience. Don't provide any descriptions or corrections for the errors".

Table 5 provides such annotation, with bold content reflecting errors in need of change, but not described in any capacity.

Table 5: : Annotated errors, with no descriptions or corrections

Your private health insurance from HUK-COBURG
Your advantages at a glance
Tailored to you - **Individual** and needs-based health insurance with **3** high-performance private health insurance **tariffs**.
Fair conditions - Permanent, guaranteed benefit promise.
Reimbursement - Currently up to 2 months' premiums in the event of no benefits*.
Our **protection** is excellent - Focus Money has once again awarded us the title of ”Fairest private health insurer” - rating ”very good” (issue 04/2024).

In the event that the translator needs hints as to the nature of the annotated errors, or, ultimately, more detailed descriptions, ChatGPT can readily provide both when prompted to do so, thereby connecting the deliberate practice conditions of opportunities to correct errors and immediate, informative feedback. The translator can make use of a prompt such as "Provide me with additional content to translate from German into English where similar errors would potentially be made" to obtain more goal-directed, pattern-specific deliberate practice. Table 6 depicts what ChatGPT generated in response to this follow-up prompt.

The content generated by ChatGPT would seem to potentially trigger similar terminology and collocation errors based on the errors produced in the previous

Table 6: : Generation of similar content in line with previous error patterns

Ihre private Rentenversicherung der HUK-COBURG
Ihre Vorteile auf einen Blick
• Individuell angepasst – Maßgeschneiderte Rentenversicherung mit flexiblen Beitragsoptionen.
• Sichere Altersvorsorge – Garantierte Rentenzahlungen ab Rentenbeginn.
• Zusätzliche Leistungen – Attraktive Zusatzleistungen wie Hinterbliebenenschutz und Berufsunfähigkeitsabsicherung.
• Ausgezeichneter Service – Kundenzufriedenheit auf höchstem Niveau, mehrfach ausgezeichnet von unabhängigen Testern.

task. Interestingly, and perhaps concerningly, the proper noun name of the same private insurance company, HUK-COBURG, is used, yet, the company itself did not author this content. ChatGPT, in generating similar content, went with the same company name, potentially crossing an ethical line in doing so. That said, and while not to downplay this ethical issue, ChatGPT is successful in providing the translator with similar tasks for purposes of error mitigation and correction practice in relation to previous performance.

6.2 6.3 Trainer presence

Somewhat inherently, ChatGPT, as an MKO, takes on the role of coach or trainer in facilitating deliberate practice. This is not to suggest that generative AI can think or train in the same capacity as a human trainer, or that human trainers become superfluous through automation. Instead, as a mechanism for self-directed learning, ChatGPT serves as a conversation partner, dependent on optimal prompting and interaction from the learner. It is also a trainer that comes with flaws, reliant on large language models and transformer input rather than years of experience in training translators or working in the language industry. As a trainer, ChatGPT may very well make mistakes, and the success of its implementation in deliberate practice contexts hinges on the learner having sufficient knowledge to detect erroneous information and question the tool accordingly.

6.3 6.4 Appropriate difficulty level

Building on the deliberate practice conditions of informative, immediate feedback and opportunities for error correction, the translator can prompt ChatGPT to provide content to be translated that is slightly more difficult in relation to previously completed tasks. This approach can address the condition of making sure work is undertaken at an appropriate difficulty level, or, ideally, at a difficulty level that is at the periphery of the translator's comfort level. The content depicted in Table 7 was generated by ChatGPT in response to the following prompt:

"Provide me with a German source text to translate in this same domain that is a little more difficult than my previous translation".

Table 7: : Generation of a slightly more difficult translation task

Ihre private Krankenversicherung der HUK-COBURG Warum sich eine private Krankenversicherung lohnt • Umfangreicher Schutz – Unsere Tarife bieten umfassende Leistungen, die weit über das hinausgehen, was gesetzliche Krankenkassen abdecken. • Schnelle Terminvergabe – Bevorzugte Behandlung bei Fachärzten und kürzere Wartezeiten. • Hohe Flexibilität – Wählen Sie aus verschiedenen Selbstbeteiligungsstufen und Tarifoptionen, die sich Ihren individuellen Bedürfnissen anpassen. • Präventionsangebote – Zugang zu exklusiven Gesundheitsprogrammen und Vorsorgeuntersuchungen. • Internationaler Schutz – Weltweite Absicherung für Reisen und Aufenthalte im Ausland. • Transparente Beiträge – Keine versteckten Kosten, klare und nachvollziehbare Prämienberechnung.

Establishing an appropriate difficulty level often proves to be a daunting task for trainers in formal translation classroom settings. It is interesting to see how ChatGPT goes about describing what exactly it is that makes this task more difficult when prompted to do so, as reflected in Table 8.

At quick glance, much of what is deemed to be "more difficult" is somewhat generic (more complex field-specific terminology, a longer text, potential lexical gaps, etc.), as opposed to personalized for the individual translator based on previous performance. This condition of deliberate practice may very well prove to be the most challenging for not only human trainers, but also ChatGPT to meet. Perhaps ChatGPT's efficacy in doing so hinges on the translator entering multiple translations as part of the prompt. These translations could then serve as a corpus of 'big data' for training the system to base what is being proposed as more difficult on concrete patterns (errors, text attributes, etc.).

Table 8: : ChatGPT's description of what makes the proposed task more difficult

This translation task is more difficult because it includes:

- Complex Vocabulary and Phrasing: The German text uses more specific and technical terms related to insurance, such as "umfangreicher Schutz," "Selbstbeteiligungsstufen," and "Prämienberechnung." Translating these terms accurately requires a good understanding of insurance terminology in both languages.
- Longer Sentences and More Detailed Content: The sentences are longer and contain more detailed information, which requires careful attention to maintain clarity and coherence in English.
- Varied Benefits and Features: The text lists multiple benefits and features of the insurance, each with its own nuances. Ensuring that each point is clearly and accurately translated without losing meaning or context adds to the complexity.
- Idiomatic Expressions: Certain phrases in German, such as "weit über das hinausgehen" and "bevorzugte Behandlung," might not have direct equivalents in English, requiring the translator to find idiomatic expressions that convey the same meaning.

6.4 6.5 Intrinsic motivation

The extent to which learners are intrinsically motivated to partake in the scenarios outlined in this chapter is intertwined with the extent to which they value and trust ChatGPT as a 'trainer' and the extent to which they embrace self-directed learning. Both can take some time and necessitate adjustment, particularly in the case of translation students who might be used to interacting with peers and instructors. As previously mentioned, intrinsic motivation in the context of self-directed, generative AI-based deliberate practice will likely stem first and foremost from the level of personalization and one-on-one interaction the technology enables. Learners are very much in the driver's seat. It is their prompting and content input that serves as a catalyst for the entire deliberate practice process. Nevertheless, intrinsic motivation could quickly falter in instances where generative AI makes errors or provides unhelpful feedback. To maintain motivation, translators would need to look beyond such isolated instances and see the bigger picture of how the technology can foster their expertise acquisition on the whole. Depending on learning styles and preferences, some learners may even find motivation at the idea of being able to openly question and challenge the non-human ‘trainer’ in such contexts, and would perhaps be less inclined to do so in contexts involving face-to-face instruction involving human trainees and trainers.

6.5 6.6 Conscious performance monitoring

Successful deliberate practice calls for the learner to be consciously engaged in reflection on their performance. In the absence of an actual human trainer and face-to-face learning, it is basically up to the learner to make sure they are on task and actively engaged in such monitoring. This important condition of deliberate practice may prove to be difficult for ChatGPT to facilitate in a fashion that’s inherently advantageous in relation to what one might find in a face-to-face learning environment. When prompted to describe ways in which it can facilitate learners’ conscious performance monitoring when utilized as a sole ‘trainer’, ChatGPT proposed such strategies as programming the tool to periodically ask users for feedback on their performance, yet also points out that it cannot track detailed usage analytics or goal progression. Instead, it can guide users on how to track their usage manually. In other words, the onus is truly on the learner when it comes to meeting this condition of deliberate practice.

7 7 Conclusion

Whereas many of the current discussions surrounding generative AI involve its potential place in translation project workflows and in the generation of translated content, this chapter advocates for leveraging its pedagogical features as a MKO and conversational agent in facilitating deliberate practice and expertise acquisition for human translators. Its primary advantages in this capacity lie in its ability to generate immediate, informative feedback in a highly personalized fashion. As far as the nature of feedback is concerned, a tool like ChatGPT is equally efficacious at annotation for purposes of error correction and description for purposes of facilitating nuanced understanding and error mitigation. It can readily provide translators with deliberate practice in relation to a given text attribute or error type, with an eye towards transferability across tasks. Personalized feedback, one-on-one training, and ongoing interaction through learner prompt entry and tool response can foster intrinsic motivation. The core deliberate practice conditions of making sure tasks are undertaken at an appropriate difficulty level and that learners engage in conscious performance monitoring, at least at the time of writing, are still somewhat of a challenge for ChatGPT, yet both can be addressed through strategic prompting.

Truly experimental studies on the benefits of self-directed, deliberate practice in relation to more traditional learning approaches are still limited in scope, for the main reason that the core conditions of deliberate practice have been quite hard to address to date (**MillerEtAl2020**:8). This is not something specific to translator training, but training in general, in all situations where learner heterogeneity, time constraints, and the absence of one-on-one training opportunities emerge as roadblocks. ChatGPT offers untapped potential to be a game changer in this regard and will hopefully draw renewed interest in both pedagogical and empirical research on deliberate practice and its place in translation expertise acquisition.